

Standardization

To convert any normal variable $X \sim N(\mu, \sigma)$ to the standard normal Z :

$$Z = \frac{X - \mu}{\sigma}$$

Use this to look up probabilities in Table A-2 or compute with technology.

Steps for Probability Questions

1. Sketch the bell curve and mark μ .
2. Plot the value(s) of interest and shade the region described.
3. Convert raw scores to z-scores using the formula above.
4. Use the z-table to find cumulative probabilities.
5. For "between" probabilities, subtract the two cumulative results.

Common Scenarios

- **Below a value:** Use the z-table value directly (area to the left).
- **Above a value:** $1 - \text{table value}$.
- **Between a and b:** $\text{Table}(b) - \text{Table}(a)$.

Example

If heights follow $\mu = 68$ in, $\sigma = 3$ in, find $P(X > 72)$.

- $z = (72 - 68)/3 \approx 1.33$.
- Table gives 0.9082 below; so above is $1 - 0.9082 = 0.0918$.

Connection to Hypothesis Testing

- Week 12 notes rely on z-table usage before moving to t-tests.
- Understanding shading and symmetry helps interpret critical regions.

Tips

- Areas are always positive; use symmetry ($P(Z < -z) = P(Z > z)$).
- Carry z to two decimals unless technology provides more precision.
- When σ is unknown and n is small, switch to the t-distribution (covered in topic-specific notes).